

Supplementary Material

S1. Derivation of transcription and translation kinetic rate expressions

We model transcription as a four-step process based on the classical work of McClure (1) and as developed previously (2). RNA polymerase (RNAP) binds the promoter of gene j to form a closed complex, which isomerizes to an open complex, from which either abortive initiation or productive elongation occurs:



where $\mathcal{G}_j$ denotes the concentration of gene j , R_X is the concentration of *free* RNAP, $(\mathcal{G}_j : R_X)_C$ and $(\mathcal{G}_j : R_X)_O$ are the closed and open complex concentrations, and m_j is the mRNA product. The key idea is that RNAP acts as a pseudo-enzyme: it binds its substrate (the gene), processes it, and dissociates, analogous to Michaelis-Menten kinetics.

Let k_+ ($\text{conc}^{-1} \text{t}^{-1}$) and k_- (t^{-1}) denote the on/off rate constants for RNAP binding at the promoter, k_I (t^{-1}) the rate constant for open complex formation, k_A (t^{-1}) the abortive initiation rate constant, and $k_{E,j}^X$ (t^{-1}) the elongation rate constant for gene j . The material balances for the closed and open complexes are:

$$\frac{d}{dt}(\mathcal{G}_j : R_X)_C = k_+ \mathcal{G}_j R_X - k_- (\mathcal{G}_j : R_X)_C - k_I (\mathcal{G}_j : R_X)_C \quad (\text{S5})$$

$$\frac{d}{dt}(\mathcal{G}_j : R_X)_O = k_I (\mathcal{G}_j : R_X)_C - k_A (\mathcal{G}_j : R_X)_O - k_{E,j}^X (\mathcal{G}_j : R_X)_O \quad (\text{S6})$$

Invoking the pseudo-steady-state approximation (justified because complex formation and dissociation are fast compared to mRNA accumulation), we set Eqns. S5–S6 to zero and solve for the steady-state complex concentrations:

$$(\mathcal{G}_j : R_X)_C = \left(\frac{k_+}{k_- + k_I} \right) \mathcal{G}_j R_X \equiv K_{X,j}^{-1} \mathcal{G}_j R_X \quad (\text{S7})$$

$$(\mathcal{G}_j : R_X)_O = \left(\frac{k_I}{k_A + k_{E,j}^X} \right) (\mathcal{G}_j : R_X)_C \equiv \tau_{X,j}^{-1} (\mathcal{G}_j : R_X)_C \quad (\text{S8})$$

where the saturation constant $K_{X,j}^{-1} \equiv k_+/(k_- + k_I)$ has units of inverse concentration and governs the affinity of RNAP for the promoter, and the dimensionless time constant $\tau_{X,j}^{-1} \equiv k_I/(k_A + k_{E,j}^X)$ compares the rate of open complex formation to the combined rates of elongation and abortive initiation. Combining these expressions, the open complex concentration is $(\mathcal{G}_j : R_X)_O = K_{X,j}^{-1} \tau_{X,j}^{-1} \mathcal{G}_j R_X$, and the rate of mRNA production for gene j is proportional to this quantity:

$$\boxed{r_{X,j} = k_{E,j}^X (\mathcal{G}_j : R_X)_O = k_{E,j}^X K_{X,j}^{-1} \tau_{X,j}^{-1} \mathcal{G}_j R_X} \quad (\text{S9})$$

where R_X is the free (unbound) RNAP concentration. The elongation rate constant $k_{E,j}^X$ is related to the polymerase elongation speed $\dot{v}_X$ (nt/s) and the gene coding length $l_{G,j}$ (nt) by $k_{E,j}^X = \dot{v}_X/l_{G,j}$, and the time constant is parameterized as $\tau_{X,j} = (k_{E,j}^X/k_I) \cdot \hat{\tau}_{X,j}$, where $k_I = 1/t_{\text{init},X}$ is the initiation rate constant estimated from McClure assays (1) and $\hat{\tau}_{X,j}$ is a dimensionless gene-specific modifier estimated from data.

An identical derivation applies to translation, with ribosome (R_L) replacing RNAP and mRNA (m_j) replacing gene ($\mathcal{G}_j$):

$$r_{L,j} = k_{E,j}^L K_{L,j}^{-1} \tau_{L,j}^{-1} m_j R_L \quad (\text{S10})$$

where $k_{E,j}^L = \dot{v}_L/l_{P,j}$ is the translation elongation rate constant, $l_{P,j}$ is the protein length in amino acids, $K_{L,j}$ is the translation saturation constant, and $\tau_{L,j} = (k_{E,j}^L/k_{I,L}) \hat{\tau}_{L,j}$ is the translation time constant.

The time constant $\tau_{X,j}$ governs whether gene j is elongation-limited or initiation-limited. Assuming abortive initiation is slow ($k_A \ll k_I, k_{E,j}^X$), then $\tau_{X,j} \simeq k_{E,j}^X/k_I$. In the initiation-limited regime ($\tau_{X,j} \gg 1$), the transcription rate reduces to the familiar Michaelis-Menten form $r_{X,j} \simeq (k_{E,j}^X/\tau_{X,j}) R_{X,T} \mathcal{G}_j / (K_{X,j} + \mathcal{G}_j)$, while in the elongation-limited regime ($\tau_{X,j} \ll 1$), the effective saturation constant is reduced by $\tau_{X,j}$, reflecting tighter RNAP binding when elongation is slow.

S2. Free machinery concentrations and competitive allocation

Consider $\mathcal{N}$ genes competing for a shared pool of RNAP with total concentration $R_{X,T}$. At any time, RNAP is either free or sequestered in closed or open complexes:

$$R_{X,T} = R_X + \sum_{j=1}^{\mathcal{N}} [(\mathcal{G}_j : R_X)_C + (\mathcal{G}_j : R_X)_O] \quad (\text{S11})$$

Substituting the steady-state expressions (Eqns. S7–S8) and factoring out R_X :

$$R_{X,T} = R_X \left[1 + \sum_{j=1}^{\mathcal{N}} \frac{(1 + \tau_{X,j}^{-1}) \mathcal{G}_j}{K_{X,j}} \right] \quad (\text{S12})$$

Solving for the free RNAP concentration yields:

$$R_X = \frac{R_{X,T}}{1 + \sum_{j=1}^{\mathcal{N}} \frac{(\tau_{X,j} + 1) \mathcal{G}_j}{\tau_{X,j} K_{X,j}}} \quad (\text{S13})$$

Each gene j sequesters RNAP in proportion to its concentration $\mathcal{G}_j$, its affinity $K_{X,j}^{-1}$, and the factor $(\tau_{X,j} + 1)/\tau_{X,j}$ which accounts for RNAP trapped in both closed and open complexes. By identical reasoning, the free ribosome concentration is:

$$R_L = \frac{R_{L,T}}{1 + \sum_{j=1}^{\mathcal{N}} \frac{(\tau_{L,j} + 1) m_j}{\tau_{L,j} K_{L,j}}} \quad (\text{S14})$$

where m_j is the mRNA concentration of gene j , which varies dynamically. Thus R_X is constant in time (since gene concentrations are fixed for linear DNA), while R_L changes as mRNA levels evolve.

For a single gene, substituting Eqn. S13 into Eqn. S9 gives $R_X = R_{X,T} \tau_{X,j} K_{X,j} / [\tau_{X,j} K_{X,j} + (\tau_{X,j} + 1) \mathcal{G}_j]$. After substitution and cancellation of $\tau_{X,j} K_{X,j}$, the transcription rate simplifies to:

$$r_{X,j} = V_{X,j}^{\max} \left(\frac{\mathcal{G}_j}{\tau_{X,j} K_{X,j} + (1 + \tau_{X,j}) \mathcal{G}_j} \right) \quad (\text{S15})$$

where $V_{X,j}^{\max} \equiv k_{E,j}^X R_{X,T} = R_{X,T} \dot{v}_X / l_{G,j}$ is the maximum transcription rate. This is the form presented in the main text.

For multiple genes, we define the allocation fraction $f_{X,j} \equiv \mathcal{G}_j / [\tau_{X,j} K_{X,j} + (1 + \tau_{X,j}) \mathcal{G}_j]$ and write the free RNAP as $R_{X,\text{free}} = R_{X,T} / (1 + \sum_j f_{X,j})$. The transcription rate for gene j in the multi-gene system is then:

$$r_{X,j} = R_{X,\text{free}} \cdot k_{E,j}^X \cdot f_{X,j} \quad (\text{S16})$$

or equivalently, $r_{X,j} = V_{X,j}^{\max} f_{X,j}$ where now $V_{X,j}^{\max} = R_{X,\text{free}} (\dot{v}_X / l_{G,j})$ uses the *free* RNAP concentration rather than the total. This formulation treats RNAP as a shared enzyme that is competitively allocated among genes, analogous to competitive substrate inhibition in enzyme kinetics. The factored form separates the global competition effect (through $R_{X,\text{free}}$, which decreases as more genes sequester RNAP) from the individual gene's kinetics (through $f_{X,j}$, which depends only on gene j 's concentration and kinetic parameters).

S3. Statistical thermodynamic derivation of the transcription control function

We model transcriptional regulation using the statistical thermodynamic framework developed by Ackers et al. (3) and adapted for synthetic circuits by Moon et al. (4). The central idea is that the promoter region of gene j can exist in a finite number of discrete microscopic configurations (microstates), each corresponding to a different combination of bound transcription factors and RNA polymerase. At thermal equilibrium, the probability of finding the promoter in microstate s is given by the Boltzmann distribution:

$$p_s = \frac{W_s g_s(\mathbf{P})}{\sum_{s' \in \mathcal{C}_j} W_{s'} g_{s'}(\mathbf{P})} \quad (\text{S17})$$

where the sum is over all possible configurations $\mathcal{C}_j$ of promoter j . Here $W_s = \exp(-\epsilon_s / k_B T)$ is the Boltzmann weight for configuration s , determined by the pseudo-energy ϵ_s relative to a reference state (the empty promoter, $\epsilon_0 = 0$, $W_0 = 1$), and $g_s(\mathbf{P})$ is a binding function that depends on the concentrations of the relevant transcriptional regulators $\mathbf{P}$. For configurations in which a regulator is bound, g_s includes a Hill-type occupancy term; for the empty or RNAP-only states, $g_s = 1$. A strongly negative ϵ_s (large W_s) indicates a thermodynamically favorable binding interaction, while a positive ϵ_s indicates an unfavorable configuration.

The transcription control function $\bar{u}_j$ is defined as the probability that the promoter is in a configuration that leads to productive transcription:

$$\bar{u}_j = \frac{\sum_{s \in \chi_j} W_s g_s(\mathbf{P})}{\sum_{s \in \mathcal{C}_j} W_s g_s(\mathbf{P})} \quad (\text{S18})$$

where $\chi_j \subseteq \mathcal{C}_j$ is the subset of transcriptionally active configurations. The value $\bar{u}_j \in [0, 1]$ represents the fraction of time the promoter spends in a productive state and directly modulates the transcription rate.

The gluconate sensor circuit contains three genes, each with a distinct promoter architecture. The P70 promoter driving GntR can exist in three states: (i) empty, (ii) core RNAP bound (without σ_{70}), or (iii) holoenzyme (σ_{70} -RNAP) bound. Both RNAP-bound states are productive, giving:

$$\bar{u}_{\text{GntR}} = \frac{W_{\text{GntR,cRNAP}} + W_{\text{GntR},\sigma_{70}} b_{\text{GntR},\sigma_{70}}}{1 + W_{\text{GntR,cRNAP}} + W_{\text{GntR},\sigma_{70}} b_{\text{GntR},\sigma_{70}}} \quad (\text{S19})$$

where $W_{\text{GntR,cRNAP}} = \exp(-\epsilon_{\text{GntR,cRNAP}}/k_B T)$ and $W_{\text{GntR},\sigma_{70}} = \exp(-\epsilon_{\text{GntR},\sigma_{70}}/k_B T)$ are Boltzmann weights for core RNAP and holoenzyme binding, respectively, and $b_{\text{GntR},\sigma_{70}} = [\sigma_{70}]^n / (K^n + [\sigma_{70}]^n)$ is a Hill function describing the fractional occupancy of σ_{70} on the holoenzyme.

The modified P70 promoter (mP70) driving Venus includes a GntR operator between the -35 and -10 regions, adding a fourth state in which the GntR repressor is bound to the operator. This state is *not* productive, so GntR occupancy reduces $\bar{u}_{\text{Venus}}$:

$$\bar{u}_{\text{Venus}} = \frac{W_{\text{Venus,cRNAP}} + W_{\text{Venus},\sigma_{70}} b_{\text{Venus},\sigma_{70}}}{1 + W_{\text{Venus,cRNAP}} + W_{\text{Venus},\sigma_{70}} b_{\text{Venus},\sigma_{70}} + W_{\text{Venus,GntR}} b_{\text{Venus,GntR}}} \quad (\text{S20})$$

D-gluconate relieves repression by binding GntR and preventing it from occupying the operator. The effective GntR concentration available for repression is $[\text{GntR}]_{\text{eff}} = (1 - f_{\text{bound}}) [\text{GntR}]_{\text{total}}$, where:

$$f_{\text{bound}} = \frac{[\text{gluconate}]^{n_{\text{gluc}}}}{[\text{gluconate}]^{n_{\text{gluc}}} + K_{\text{gluc}}^{n_{\text{gluc}}}} \quad (\text{S21})$$

is the fraction of GntR bound to D-gluconate (and therefore unable to repress), K_{gluc} is the dissociation constant, and n_{gluc} is the Hill coefficient for cooperative gluconate binding to the GntR dimer. At saturating gluconate ($f_{\text{bound}} \rightarrow 1$), repression is fully relieved. Finally, since σ_{70} is supplied exogenously as protein and is not encoded on a gene in the circuit, its constitutive promoter control function is $\bar{u}_{\sigma_{70}} = W_{\sigma_{70},\text{RNAP}} / (1 + W_{\sigma_{70},\text{RNAP}})$ with $W_{\sigma_{70},\text{RNAP}} = \exp(-0.5)$ fixed.

The partition function formalism provides a principled bridge between molecular-level binding events and macroscopic gene expression rates. Each pseudo-energy ϵ_s encodes the net free energy change for a specific promoter configuration relative to the empty state,

integrating contributions from protein-DNA contacts, conformational changes, and steric effects. The resulting control function $\bar{u}_j$ is a smooth, bounded function that modulates the kinetic rate $r_{X,j}$ at each time step, coupling the thermodynamic state of the promoter to the dynamic mass balance equations.

S4. Consumable resource depletion

A distinguishing feature of cell-free systems is that the small-molecule substrates required for transcription and translation — nucleoside triphosphates (NTPs), energy cofactors such as creatine phosphate, and amino acids — are present in finite supply and are not regenerated by cellular metabolism. As these consumable resources are depleted over the course of the reaction, the rates of transcription and translation decline, eventually halting gene expression entirely. Previous cell-free models have accounted for this decline using empirical time-dependent decay functions, such as logistic or exponential terms with predetermined onset times and half-lives (5; 6). While these functions can reproduce the observed saturation of mRNA and protein trajectories, they are ad hoc: the decay rate is a fixed function of time rather than a consequence of the actual biosynthetic load placed on the system. This means they cannot predict how resource depletion would change if, for example, additional genes were added to the circuit or promoter strengths were altered.

To address this limitation, we formulated a mechanistic resource depletion model in which the rate of resource consumption is driven by the cumulative biosynthetic flux. We define two dimensionless state variables: $\epsilon_X(t) \in [0, 1]$, representing the fractional availability of consumable resources that sustain transcription, and $\epsilon_L(t) \in [0, 1]$, representing the fractional availability of consumable resources that sustain translation, with $\epsilon_X(0) = \epsilon_L(0) = 1$. The depletion dynamics are governed by:

$$\frac{d\epsilon_X}{dt} = -\alpha_X \left(\sum_{j=1}^{\mathcal{N}} l_{G,j} r_{X,j} \bar{u}_j \right) \epsilon_X \quad (\text{S22})$$

$$\frac{d\epsilon_L}{dt} = -\alpha_L \left(\sum_{j=1}^{\mathcal{N}} l_{P,j} r_{L,j} \right) \epsilon_L \quad (\text{S23})$$

where α_X and α_L are consumption rate constants with units of inverse flux, $l_{G,j}$ is the coding length of gene j in nucleotides, and $l_{P,j}$ is the length of protein j in amino acids. The summation terms represent the total biosynthetic flux: each gene's transcription rate $r_{X,j} \bar{u}_j$ is weighted by its coding length $l_{G,j}$, reflecting the stoichiometric requirement that longer transcripts consume more NTPs per molecule synthesized. Likewise, each translation rate $r_{L,j}$ is weighted by the protein length $l_{P,j}$, since longer polypeptides consume more amino acids and GTP equivalents per molecule.

The multiplicative structure of Eqns. S22–S23 creates a natural feedback: as ϵ_X decreases, it attenuates the mRNA production terms in the balance equations (Eqn. S24), which reduces the transcription flux, which in turn slows further depletion of ϵ_X . This self-limiting behavior produces the characteristic sigmoidal-to-saturating protein accumulation curves observed in cell-free reactions without requiring an externally imposed decay schedule. Importantly, the depletion rate depends on the *actual* biosynthetic activity in the system: circuits with more

genes, stronger promoters, or longer coding sequences will deplete resources faster, while less active circuits will deplete them more slowly. This load-dependent behavior is what distinguishes the mechanistic formulation from empirical decay functions and enables the model to generalize across conditions with different gene dosages or circuit architectures.

The separation of resource depletion into distinct transcriptional (ϵ_X) and translational (ϵ_L) pools is motivated by the observation that these processes consume different substrates: transcription primarily consumes NTPs and the energy cofactors that regenerate them, while translation primarily consumes amino acids, charged tRNAs, and GTP. By tracking these pools independently, the model can identify which resource class is limiting under a given set of conditions — a capability that proved central to the bottleneck analysis presented in the main text.

S5. Complete model equations

The full model consists of 11 ordinary differential equations. Gene concentrations are constant (linear DNA, no replication): $d\mathcal{G}_j/dt = 0$ for $j = 1, 2, 3$. The mRNA balances couple transcription (modulated by promoter control and resource availability) with first-order degradation:

$$\frac{dm_j}{dt} = r_{X,j} \bar{u}_j \epsilon_X - \theta_{m,j} m_j \quad j = 1, 2, 3 \quad (\text{S24})$$

where $\theta_{m,j} = \delta_{m,j} \ln 2/t_{1/2,m}$ is the effective degradation rate constant, $\delta_{m,j}$ is a gene-specific modifier, and $t_{1/2,m}$ is the characteristic mRNA half-life. The protein balances have an analogous structure:

$$\frac{dp_j}{dt} = r_{L,j} \epsilon_L - \theta_{p,j} p_j \quad j = 1, 2, 3 \quad (\text{S25})$$

where $\theta_{p,j} = \delta_{p,j} \ln 2/t_{1/2,p}$. Together with the resource depletion equations (Eqns. S22–S23), the algebraic expressions for transcription control (Eqns. S19–S20), machinery allocation (Eqns. S13–S14), and kinetic rates (Eqns. S9, S10), this defines a closed system of 11 ODEs with 25 estimated parameters.

References

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